TED University | Department of Computer Engineering

Project Proposal

Design and Development of a Smart Cane Attachment with Obstacle Detection and User Alerting for Visually Impaired Individuals

CMPE 491 Senior Design Project I | Group 18 | 16 October 2026

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Project website: https://mertsaglm.github.io/tedu-smart-cane/

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Project description

We plan to develop a detachable electronic attachment for existing white canes to help visually impaired individuals notice obstacles in their path, both at ground level and at different heights. During a visit to an association for people with visual impairments, our team learned that obstacles above the ground, such as signs and tree branches, were a particular difficulty. This helped us understand the need for a device that supports users in different everyday situations. The attachment will measure the distance to nearby obstacles and warn the user through vibration and/or sound, with stronger or more frequent feedback as an obstacle gets closer. Our aim is to make the device affordable, easy to attach and comfortable to use alongside the white cane.

We will start with a simple prototype that detects nearby obstacles and gives clear distance-based alerts. We plan to compare ultrasonic and time-of-flight sensors and test suitable sensor positions for obstacles at different heights, including near the ground. We will also explore how measurements of cane movement can help reduce unnecessary alerts. The hardware and feedback method will be chosen through practical tests and user feedback. After testing the basic prototype, we will add further features step by step, such as recognising obstacle types and providing more detailed information about the surroundings. We will check detection performance, missed obstacles, false alerts, response time, weight, battery use and attachment stability. Our initial target for prototype materials is around TRY 10,000; the actual cost will depend on component choices and prices. In CMPE 491, we will work on requirements, analysis, high-level design and an initial demonstration. Development and testing will continue in CMPE 492.

Initial literature review

We reviewed existing products and research projects to understand what has already been done. WeWALK combines upper-body obstacle detection with navigation and voice assistance [1]. GoSense Rango is a detachable white-cane module that communicates obstacle position through spatial audio [2]. SmartCane, developed at IIT Delhi with its partners, uses ultrasonic sensors and vibration patterns that change with distance [3]. These examples show that attaching sensors to a cane is already a well-established idea. For our project, we will focus on a low-cost, detachable design and useful alerts while the cane is moving.

Other projects include more advanced sensing and guidance. Glidance is developing the wheeled Glide mobility aid, which provides physical steering [4]. Stanford researchers demonstrated an augmented cane with robotic sensing and steering [5]. In China, Mai et al. combined 2D LiDAR, an RGB-D camera and object recognition in a wheeled smart cane [6]. These projects offer useful ideas for later development, but their additional hardware and processing needs are also important considerations for a small attachment.

Based on this review, we plan to begin with basic obstacle detection and distance-based alerts, and build on the prototype as testing progresses. Our tests will include obstacles near the ground and at different heights. We will compare a simple distance-threshold approach with one that also considers cane movement, and look at how clearly users can understand the feedback. Product descriptions help us understand the features offered by existing devices, but we will need our own tests to evaluate the performance of our prototype.

References

[1] WeWALK. Smart Cane 2 product overview. Accessed 16 October 2026.

[2] GoSense. Rango product sheet. Accessed 16 October 2026.

[3] Saksham Trust. SmartCane overview. Accessed 16 October 2026.

[4] Glidance. Glide product overview. Accessed 16 October 2026.

[5] Stanford University School of Engineering. A team of researchers build a $400 self-navigating smart cane, 2021. Accessed 16 October 2026.

[6] C. Mai et al. A Smart Cane Based on 2D LiDAR and RGB-D Camera Sensor-Realizing Navigation and Obstacle Recognition. Sensors, 24(3), 870, 2024. DOI: 10.3390/s24030870. Accessed 16 October 2026.